

Companion LXII: Betti Numbers of Weak-Lensing Fields in the Relaxation-Driven Cyclic Cosmology

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Abstract

Betti numbers provide a topological characterization of cosmological fields by counting connected components, loops, and void-like structures in excursion sets. In weak-lensing analyses, they offer a powerful probe of non-Gaussianity, connectivity, and the topology of the convergence field. In the standard cosmological model, Betti numbers reflect the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the connectivity, loop structure, and void topology of κ -fields in a characteristic way. This Companion develops a continuous description of Betti numbers in RDCC, deriving how the relaxation scale influences the homology of excursion sets and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive topological probe of the RDCC gravitational field.

1 Introduction

Weak-lensing convergence maps contain rich topological information. While Minkowski functionals describe morphology and global topology, Betti numbers provide a more refined homological description by counting connected components, loops, and voids in excursion sets.

In the standard cosmological model, Betti numbers reflect the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the connectivity, loop structure, and void topology of κ -fields in a scale-dependent manner, producing a distinctive signature in Betti-number analyses.

2 Betti Numbers of the Convergence Field

For a 2D field such as the convergence $\kappa(\hat{\mathbf{n}})$, the Betti numbers of an excursion set $\kappa > \nu\sigma_\kappa$ are:

$$\beta_0(\nu) = \text{number of connected high-}\kappa \text{ regions}, \quad (1)$$

$$\beta_1(\nu) = \text{number of loops or holes in the excursion set}. \quad (2)$$

These quantities encode:

- the fragmentation of high- κ structures,
- the presence of ring-like or loop-like features,
- the topology of voids and ridges.

3 RDCC Modification of the Convergence Field

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (3)$$

The variance becomes

$$\sigma_{\kappa, \text{RDCC}}^2 = \sigma_{\kappa}^2 \left[1 - \chi_{\sigma} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (4)$$

These modifications alter the topology of excursion sets.

4 Betti Numbers in RDCC

The RDCC Betti numbers become

$$\beta_{0, \text{RDCC}}(\nu) = \beta_0(\nu) \left[1 - \chi_0 \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right], \quad (5)$$

$$\beta_{1, \text{RDCC}}(\nu) = \beta_1(\nu) \left[1 - \chi_1 \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (6)$$

These modifications reflect the relaxation-driven changes in connectivity and loop structure.

5 Topological Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Betti numbers therefore exhibit:

- fewer small-scale connected components (β_0 suppression),
- fewer small-scale loops (β_1 suppression),
- larger, smoother connected high- χ regions,
- a shift in the β_0 - β_1 crossing point,
- a characteristic turnover near ℓ_{rel} ,
- modified topology of voids and ridges.

These signatures provide a direct observational probe of the RDCC gravitational field in topological space.

6 Conclusion

Betti numbers of weak-lensing fields in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the connectivity, loop structure, and void topology of χ -fields in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of topological cosmology, nonlinear structure, and the interplay between light and gravity in RDCC.

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